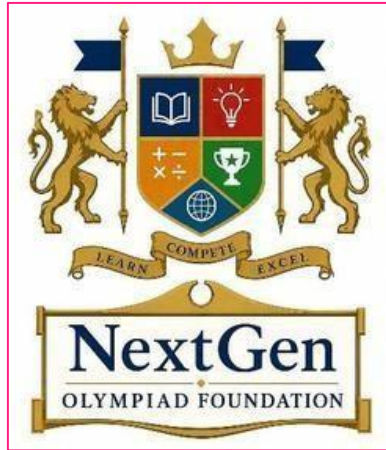




GRADE 2

Computational Thinking

National Level Examination (NLE)

SUBJECT CODE

Questions: 30

Total Marks: 100

Time: 45 Minutes

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- **CAREFULLY CHECK YOUR NAME, CLASS, ROLL NUMBER, AND OTHER REQUIRED DETAILS ON THE OMR ANSWER SHEET.**
- **DO NOT FORGET TO SIGN THE OMR SHEET. ALSO WRITE YOUR ROLL NO. ON THE QUESTION PAPER BOOKLET.**
- **USE ONLY BLACK/BLUE BALL PEN AND HB PENCIL TO FILL THE OMR ANSWER SHEET.**
- **ALL QUESTIONS ARE COMPULSORY AND CARRY ONE MARK EACH.**
- **SECTION A: LOGICAL AND ANALYTICAL REASONING (9 QUESTIONS); SECTION B: PATTERN SEQUENCE AND CODING (11 QUESTIONS); SECTION C: ALGORITHM AND STEP BY STEP THINKING (6 QUESTIONS); SECTION D: ALGORITHMIC THINKING & LOGICAL SEQUENCING (6 QUESTIONS); SECTION E: COMPUTATIONAL THINKING AND ROBOT COMMANDS (4 QUESTIONS).**
- **ONLY OMR ANSWER SHEETS WILL BE EVALUATED.**
- **NO NEGATIVE MARKING FOR INCORRECT ANSWERS.**
- **YOU MAY USE THE SPACE PROVIDED ON THE LAST PAGE FOR ROUGH WORK (IF NEEDED).**
- **DIGITAL DEVICES LIKE CALCULATORS, SMART WATCHES, OR MOBILE PHONES ARE NOT ALLOWED.**
- **AFTER THE EXAM, SUBMIT BOTH THE OMR SHEET AND THE QUESTION PAPER TO THE INVIGILATOR.**

SECTION A – LOGICAL & ANALYTICAL REASONING

Q1. If 20 years ago, a man was 10 years old, how old is he now?

- a) 20 years
- b) 40 years
- c) 30 years
- d) 10 years

Q2. Four children stand in a line. Riya is before Rohan. Rohan is before Aman. Aman is before Simran. Who is third?

- a) Riya
- b) Rohan
- c) Aman
- d) Simran

Q3. A robot is facing North. It turns right. Which direction is it facing?

- a) West
- b) South
- c) East
- d) North

Q4. Rahul is older than Sara. Sara is older than Kanika. Who is the oldest?

- a) Sara
- b) Rahul
- c) Kanika
- d) All of them are same age

Q5. The players of a cricket team have their matches on Tuesday and Wednesday. It is Saturday today. After how many days will they have their first match?

- a) 2 days
- b) 4 days
- c) 3 days
- d) 1 day

Q6. Mena has to reach the railway station at 6:30 am. She lives at a distance of 2 hours from the station. At what time should she leave home to reach her destination?

- a) 4:00 am
- b) 3:00 am
- c) 4:30 pm
- d) 4:30 am

Q7. Few friends are sitting in a row in the sequence given below: Riya, Siya, Dhruv, Megha, Mohit, Rahul, Aman. Who is sitting at the centre?

- a) Mohit
- b) Megha
- c) Dhruv
- d) Megha and Mohit

Q8. A bus has 25 passengers. At the first stop, 7 get down and 5 get in. At the second stop, 4 get down. How many passengers are now on the bus?

- a) 17
- b) 19
- c) 21
- d) 23

Q9. A room has 3 switches. Only one switch turns on the fan. Switch A is not the fan switch. Switch B is not the fan switch. There is only one fan switch. Which switch is it?

- a) A
- b) B
- c) C
- d) Cannot be found

SECTION B – PATTERNS, SEQUENCES & CODING

Q10. If A = 1, B = 2, C = 3, then what is the value of CAB?

- a) 5
- b) 6
- c) 7
- d) 8

Q11. If A = 1, B = 2, C = 3, then what is the value of CAB?

- a) 5
- b) 6
- c) 7
- d) 8

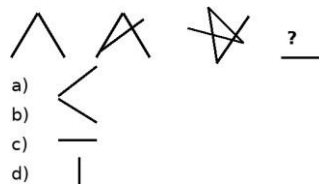
Q12. Find the missing step: 10 → 20 → 40 → 80 → _____.

- a) 100
- b) 120
- c) 140
- d) 160

Q13. Complete the pattern: AB, BCD, CDEF, DEFGH, _____.

- a) EFGHI
- b) EFGHIJ
- c) FGHIJ
- d) DEFGHI

Q16. Complete the pattern by drawing a correct line.



Q17. If APPLE → 1-16-16-12-5, then MANGO will be:

- a) 13-1-14-7-15
- b) 12-1-14-6-15
- c) 13-2-14-7-15
- d) 14-1-13-7-15

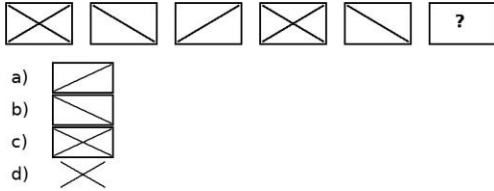
Q18. Find the missing numbers: 9, 18, _____, 36, 45, 54, _____.

- a) 27, 62
- b) 27, 64
- c) 47, 27
- d) 27, 63

Q14. Which one number in the given pattern is not correct? $10 \rightarrow 22 \rightarrow 44 \rightarrow 94 \rightarrow 190$

- a) 22
- b) 04
- c) 190
- d) 10

Q15. Look at the pattern and complete the given sequence.



Q19. Find the missing number: 2, 4, 16, 32, _____.

- a) 42
- b) 62
- c) 64
- d) 60

Q20. Fill the missing number: 100, 90, 85, 75, 70, _____.

- a) 65
- b) 60
- c) 80
- d) 85

SECTION C – ALGORITHMIC & STEP-BY-STEP THINKING

Q21. Which instruction is unnecessary when teaching a robot to make lemonade?

- a) Take a glass
- b) Add water
- c) Add lemon juice
- d) Wear blue shoes

Q24. Which sequence is in the correct order?

- a) Wake up $\rightarrow$ Sleep $\rightarrow$ Brush $\rightarrow$ Eat
- b) Brush $\rightarrow$ Wake up $\rightarrow$ Eat $\rightarrow$ Sleep
- c) Wake up $\rightarrow$ Brush $\rightarrow$ Eat $\rightarrow$ Go to school
- d) Eat $\rightarrow$ Sleep $\rightarrow$ Wake up $\rightarrow$ Brush

Q22. Which instruction should come first when making a sandwich?

- a) Eat the sandwich
- b) Put filling on the bread
- c) Take the bread
- d) Cut the sandwich

Q25. Look at the pairs: $2 \rightarrow 6$, $3 \rightarrow 9$, $4 \rightarrow 12$, $5 \rightarrow 15$. Which rule is being used?

- a) Add 4
- b) Multiply by 2
- c) Multiply by 3
- d) Add 3

Q23. A number machine does $\times 2$ and then $+1$. If the input is 4, what was the output?

- a) 7
- b) 8
- c) 9
- d) 10

Q26. A computer follows this rule: If the number is even $\rightarrow$ add 5. If the number is odd $\rightarrow$ add 2. What will happen to 8?

- a) 10
- b) 12
- c) 13
- d) 15

SECTION D – COMPUTATIONAL THINKING & ROBOT COMMANDS

Q27. A robot starts with A. It follows the commands shown below. Where does it finish? $\rightarrow \rightarrow \downarrow \rightarrow \uparrow \rightarrow$

- a) B
- b) C
- c) D
- d) E

Q28. A robot has four buttons: A = $+2$, B = $\times 2$, C = -1 . Starting with 5, which sequence gives 13?

- a) B $\rightarrow$ C $\rightarrow$ A
- b) A $\rightarrow$ B $\rightarrow$ C
- c) C $\rightarrow$ B $\rightarrow$ A
- d) A $\rightarrow$ C $\rightarrow$ B

Q29. A robot has 3 commands: $\triangle$ = move forward, $\circ$ = turn right, $\star$ = move backward. It follows: $\circ \triangle \triangle \star$. How many times does it move forward?

- a) 2
- b) 3
- c) 4
- d) 5

Q30. Which is the same as $\circ \triangle \blacksquare$ when read backwards?

- a) $\circ \triangle \blacksquare$
- b) $\circ \blacksquare \triangle$
- c) $\circ \blacksquare \triangle$
- d) $\circ \blacksquare \triangle$